

Experiment-7

Topic: Database Normalization – Functional Dependencies, Candidate Keys and Normal Forms

Q1: Relation: R(A, B, C, D)

Functional Dependencies:

- $AB \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow A$

Q2: Relation: R(A, B, C, D, E)

Functional Dependencies:

- $A \rightarrow D$
- $B \rightarrow A$
- $BC \rightarrow D$
- $AC \rightarrow BE$

Q3: Relation: R(A, B, C, D, E)

Functional Dependencies:

- $B \rightarrow A$
- $A \rightarrow C$
- $BC \rightarrow D$
- $AC \rightarrow BE$

Q4: Relation: R(A, B, C, D, E, F)

Functional Dependencies:

- $A \rightarrow BCD$
- $BC \rightarrow DE$
- $B \rightarrow D$
- $D \rightarrow A$

Tools / Platform Used:

- Windows
- PostgreSQL
- PgAdmin

Solutions:

Q1.

$(AB)^+ = ABCD \rightarrow AB$ is a candidate key.

$(BC)^+ = BC \rightarrow D \rightarrow A \rightarrow BC$ is a candidate key.

$(BD)^+ = BD \rightarrow A \rightarrow C \rightarrow BD$ is a candidate key.

Therefore:

- Candidate Keys = {AB, BC, BD}
- Prime Attributes = {A, B, C, D}
- Non-Prime Attributes = { }

Q2.

$(AC)^+ = AC \rightarrow D, B, E \rightarrow AC$ is a candidate key.

$(BC)^+ = BC \rightarrow A \rightarrow D$ and $AC \rightarrow E \rightarrow BC$ is a candidate key.

Therefore:

- Candidate Keys = {AC, BC}
- Prime Attributes = {A, B, C}
- Non-Prime Attributes = {D, E}

Q3.

$A^+ = A \rightarrow C; AC \rightarrow B, E; BC \rightarrow D$

Therefore $A^+ = ABCDE$, so A is a candidate key.

$B^+ = B \rightarrow A \rightarrow C; BC \rightarrow D; AC \rightarrow E$

Therefore $B^+ = ABCDE$, so B is a candidate key.

Therefore:

- Candidate Keys = {A, B}
- Prime Attributes = {A, B}
- Non-Prime Attributes = {C, D, E}

Q4.

$A^+ = ABCDE$, but F is not obtained. Therefore, A is not a key by itself.

$AF^+ = ABCDEF \rightarrow AF$ is a candidate key.

$BF^+ = ABCDEF$ because $B \rightarrow D \rightarrow A$ and $A \rightarrow C$; therefore, BF is a candidate key.

$DF^+ = ABCDEF$ because $D \rightarrow A \rightarrow B, C$ and $BC \rightarrow E$; therefore, DF is a candidate key.

Therefore:

- Candidate Keys = {AF, BF, DF}
- Prime Attributes = {A, B, D, F}
- Non-Prime Attributes = {C, E}

Learning Outcomes:

- Practiced PostgreSQL concepts of Normalisation and functional dependencies.